

Integrating Wearable IMU Sensors with an OpenSim Biomechanical Model to Collect and Analyze Human Movement Outside the Lab

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Background

- There are many approaches when it comes to analyzing human movement. Quantitative examples include; optical motion capture and Inertial Measuring Unit (IMU) sensors, while the Functional Movement Screen (FMS) is a qualitative clinical technique.

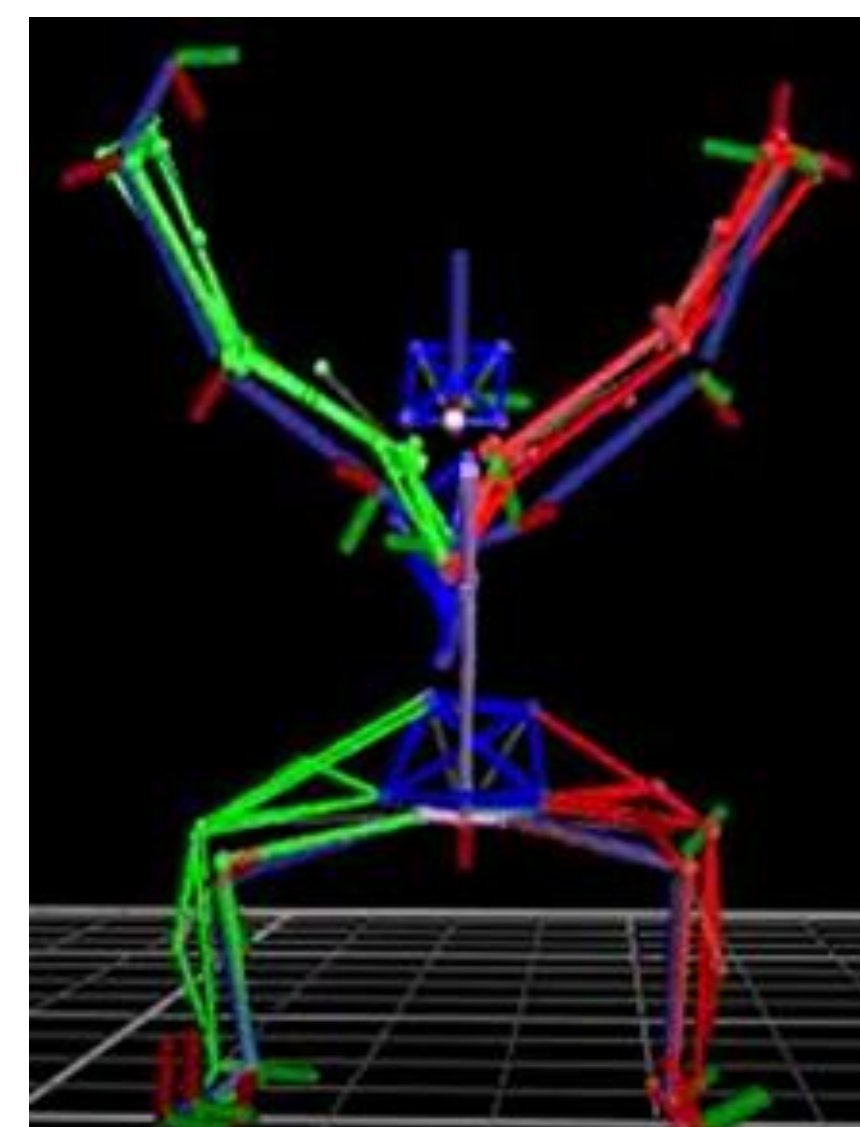


EBL Vicon Motion Capture

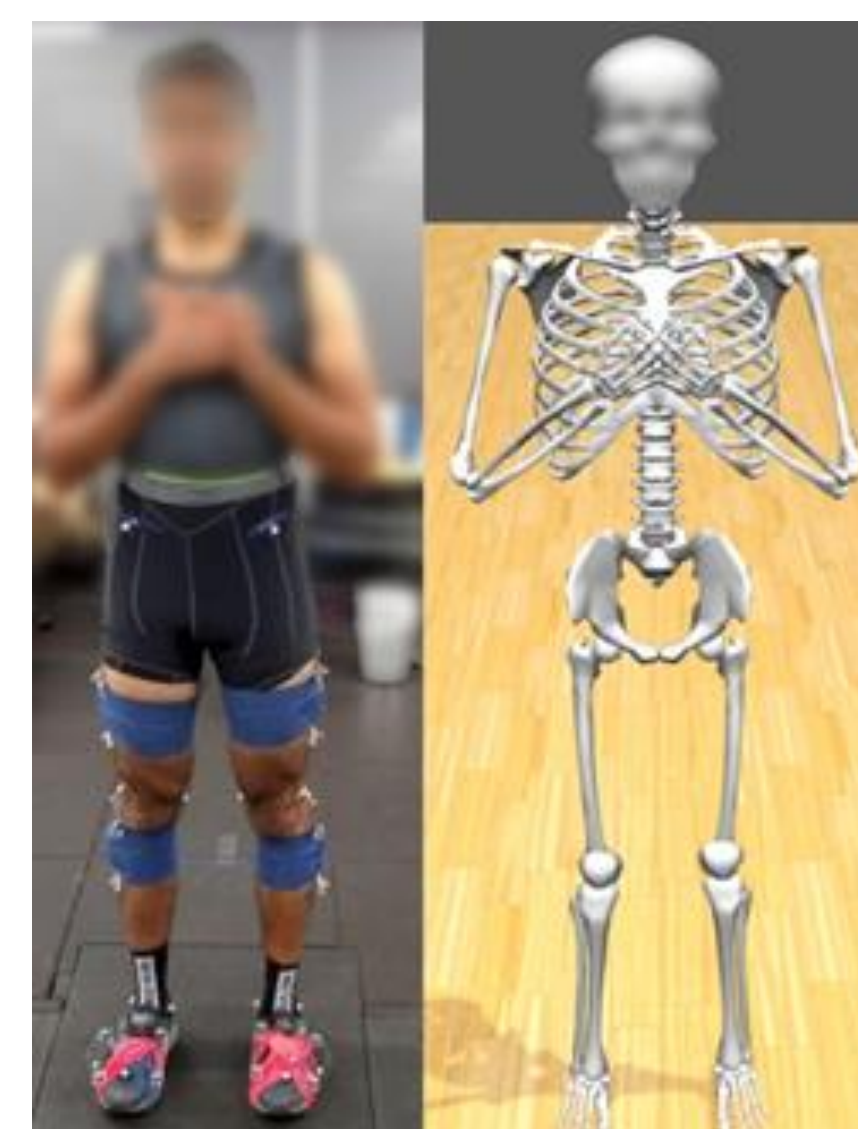
Wearable IMU sensors

7 movements of the Functional Movement Screen

- Optical motion capture is the current gold standard for biomechanical analysis. However, many researchers, coaches and trainers do not have the access, time, and/or expertise to assess subjects in a motion capture laboratory (1). IMU sensors are a cheaper and possibly more convenient way to quantify human motions during real world activities (2).



Vicon Plug-In-Gait biomechanical model



Opensim biomechanical model

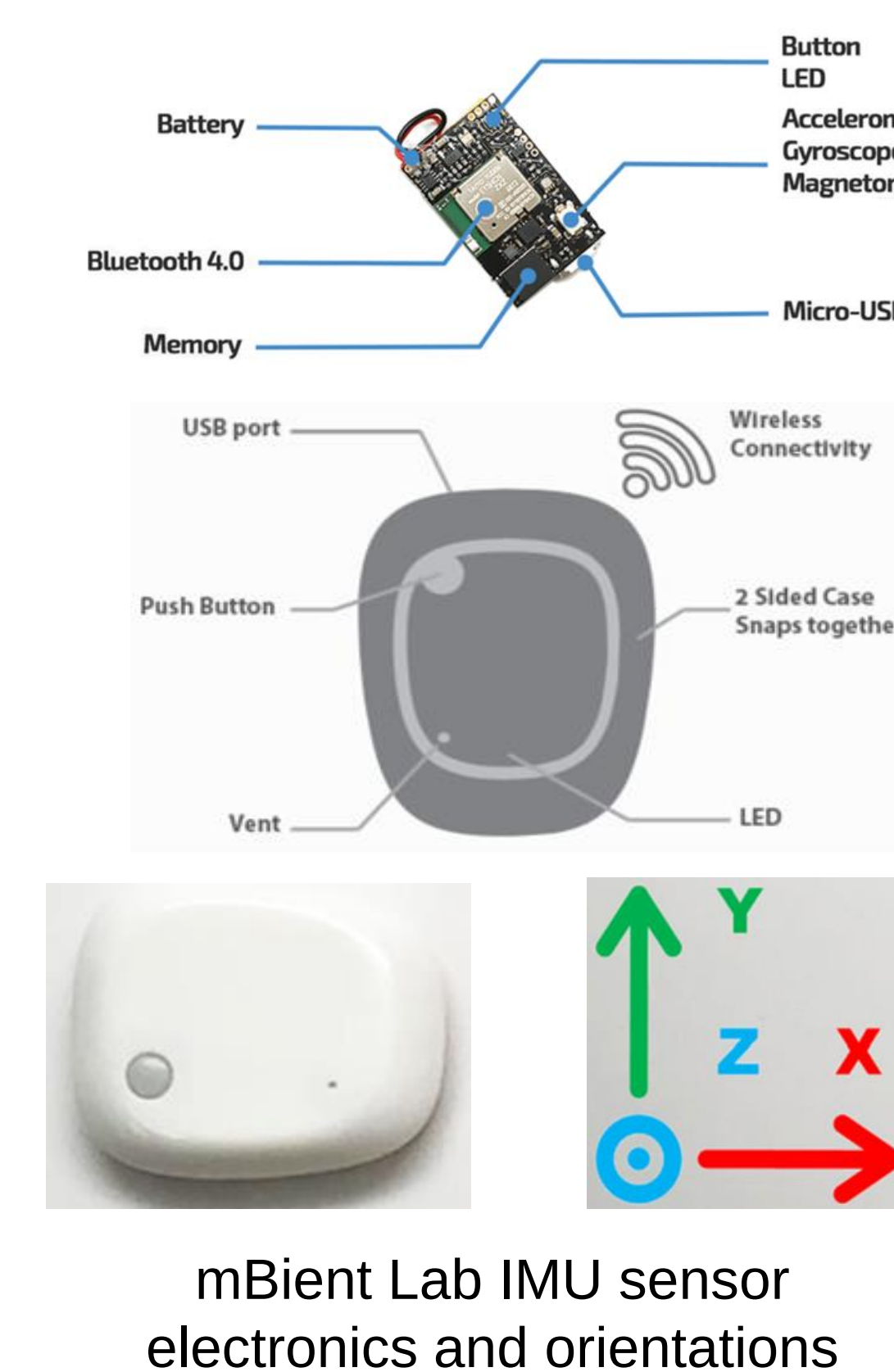
Objective

- The objective of this research is to integrate wireless wearable IMU sensors with OpenSim software (3) to collect and analyze human movement. This method will be validated compared to motion capture. Then, real world human testing will compare quantitative IMU kinematics with qualitative FMS assessment by human movement professionals.

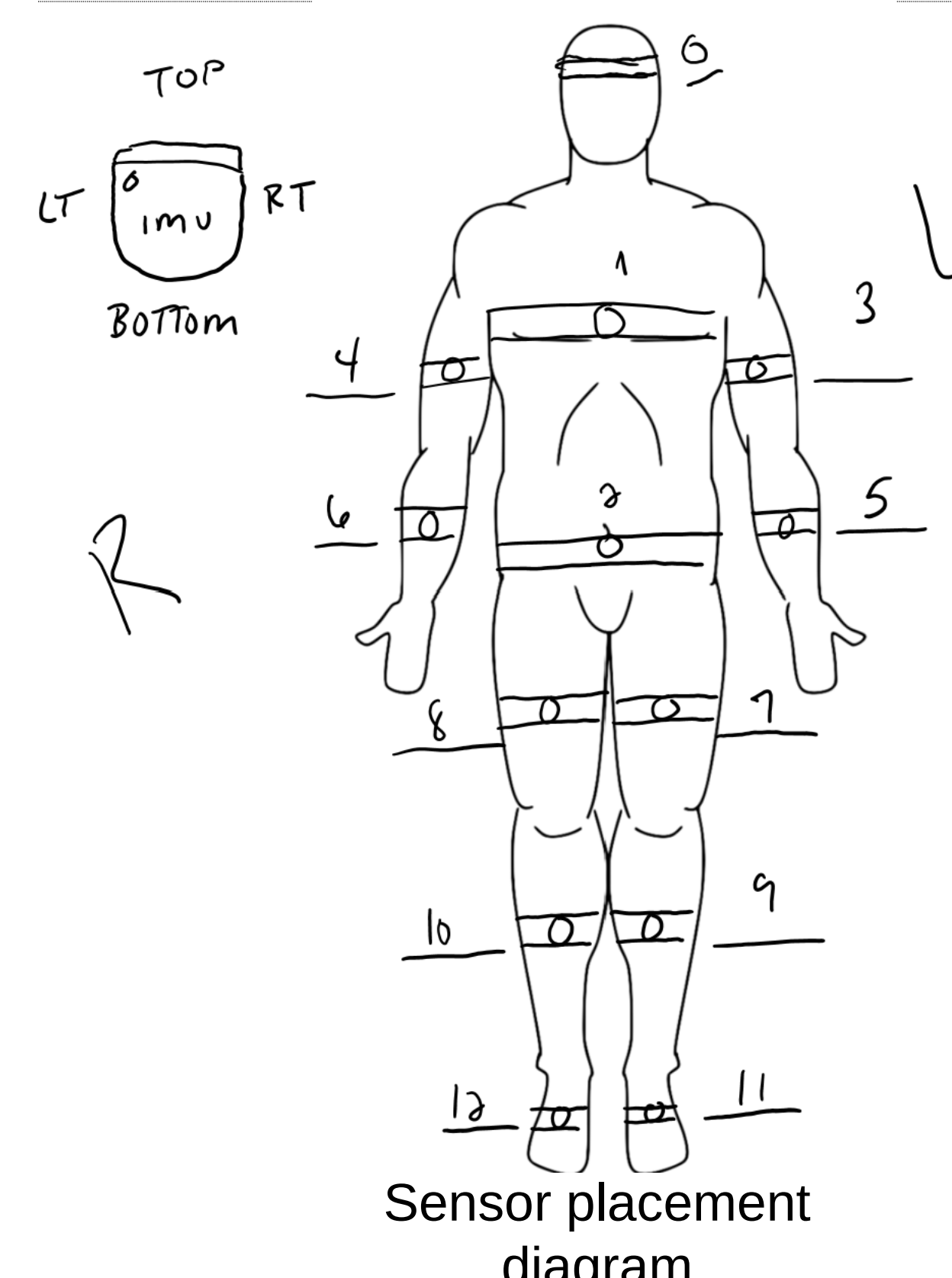
References : (1) Dowling et al. Inertial Sensor Based Feedback Can Reduce Key Risk Metrics for Anterior Cruciate Ligament Injury During Jump Landing, 2012; (2) Wu et al. Development of an Automatic Functional Movement Screening System with IMU Sensors, 2021; (3) Delp et al. OpenSim: Open-source Software to Create and Analyze Dynamic Simulations of Movement, 2007; (4) Mjosund et al. Clinically Acceptable Agreement Between the ViMove Wireless Motion Sensor System and the Vicon Motion Capture When Measuring Lumbar Region Inclination Motion in the Sagittal and Coronal Planes (2017)

Materials and Methods

- Standard Vicon procedure:** attach markers, calibrate system, T-pose, capture dynamic trials, analyze using Plug-In-Gait model tools, export .csv file for data analysis of joint kinematics.
- New IMU procedure:** attach sensors, start recording, T-pose and FMS movements, stop recording, process data files to OpenSim format, place IMUs on model, analyze using IMU inverse kinematics tools, obtain graphs of joint kinematics.



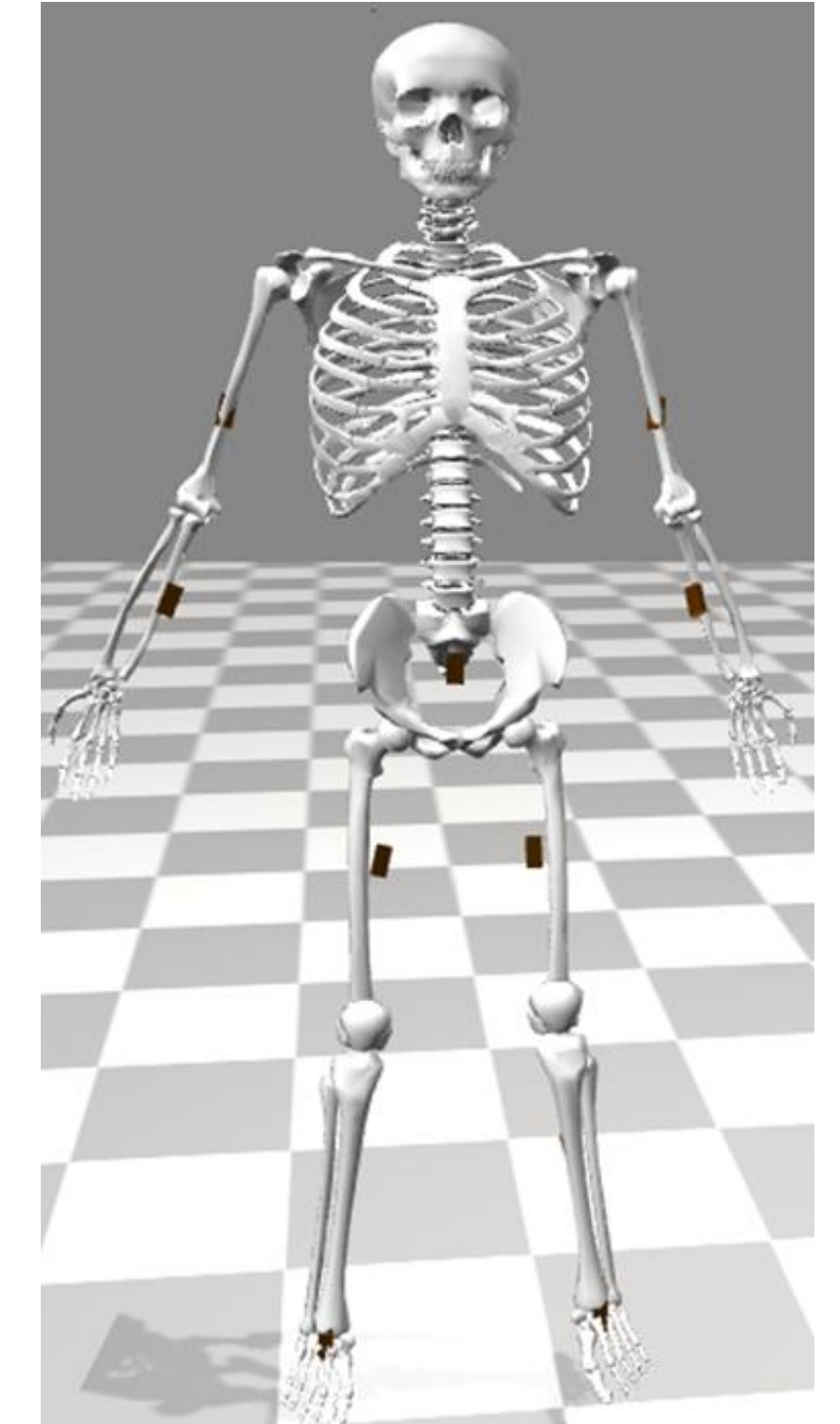
mBient Lab IMU sensor electronics and orientations



Sensor placement diagram



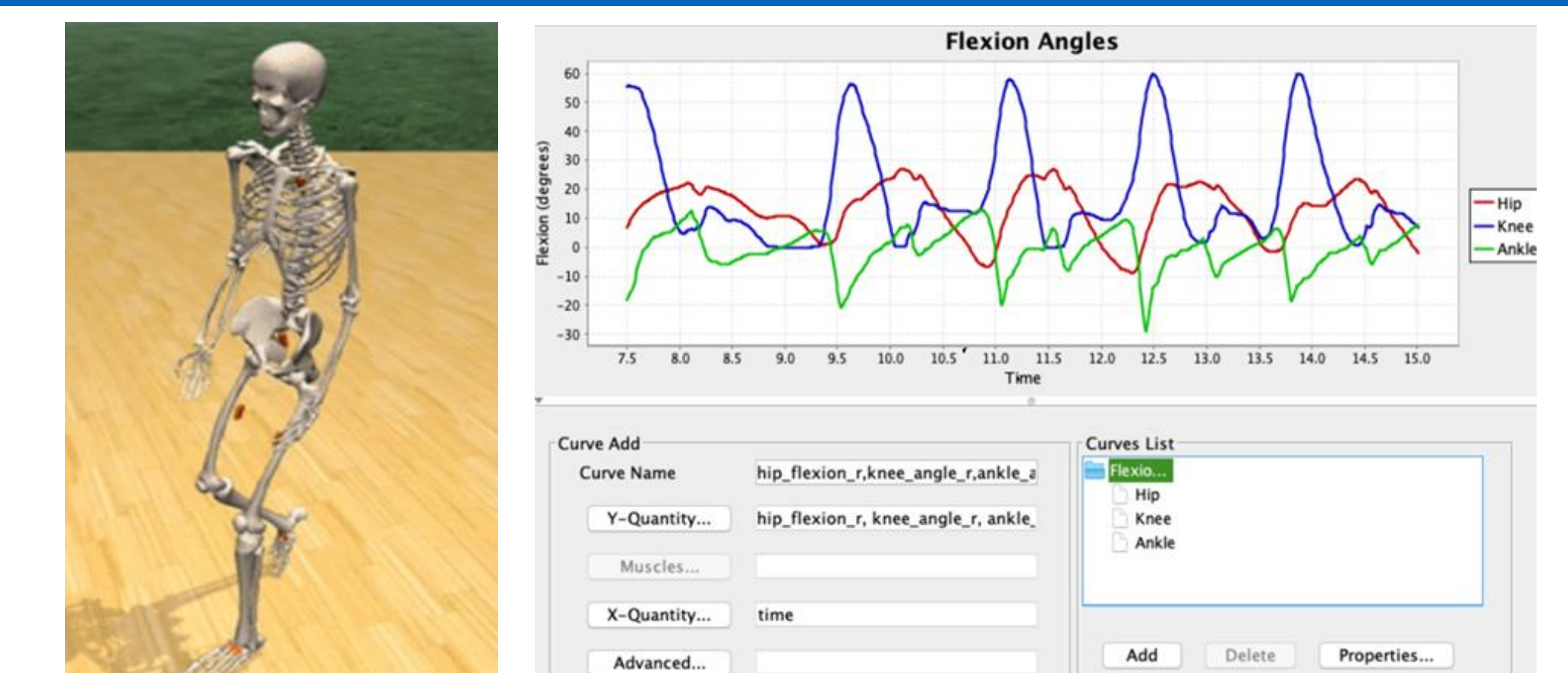
Subject T-pose for placement calibration



Opensim biomechanical model with IMU sensors

Results

- The kinematic parameters of interest include hip adduction and abduction angle, hip flexion angle, knee flexion and extension angle, shoulder range or motion ROM, ankle dorsiflexion, and trunk relative angle. Strong similarities between IMU and marker based kinematics are expected.



Representative flexion/extension kinematics obtained from IMU sensors and processed in OpenSim

Discussion

- IMU sensors do not require "line of sight" so FMS movements that are performed on the ground can be more easily tracked. They are also portable, to track motions in the clinic or during practices and games. Future research should attempt to collect data using a bigger sample size with diverse populations. Other movements, such as activities of daily living (ADL) and highly dynamic movements such as sprinting, jumping and sports could also be studied (4). Sensor drift can be an issue with IMU data, and GPS or other types of sensors would also be necessary to track the position.

